

CHAPTER 24

James Webb Space Telescope ACS design

This chapter describes a design for the James Webb Space Telescope ACS. The great thing about NASA and ESA programs is that the engineers write numerous papers. These allow you to learn about spacecraft design. This chapter does not present the actual design used on the spacecraft, just a design that meets some of the published requirements. The design in this chapter is unique to this book. It also relies entirely on information available to the public. There are always multiple ways to design a spacecraft-control system.

The James Webb is an inertial precision-pointing spacecraft. It needs to point with extreme accuracy and with very low jitter at distant stars. The telescope is shown in Fig. 24.1.



Figure 24.1 James Webb Space Telescope. Image courtesy of NASA.

The structure of the control system is shown in Fig. 24.2. There are two control functions, momentum management and attitude control. Both produce torque-demand vectors. These are converted into thruster pulsewidth and reaction-wheel commands using linear programming. Sensing is done via star cameras and inertial measurement units (IMUs). The attitude determination system uses an unscented Kalman filter with the IMU as the dynamical base.

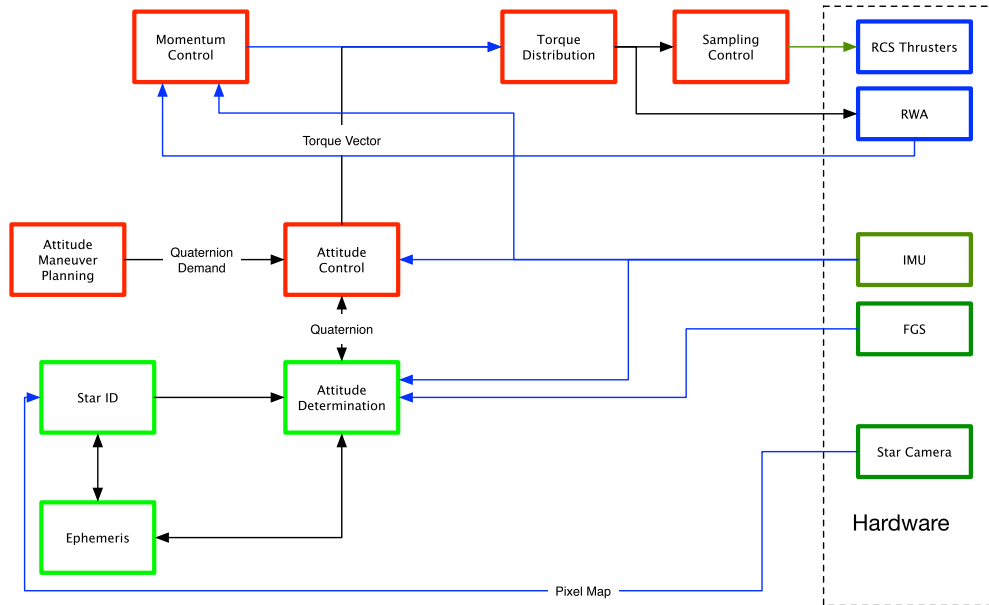


Figure 24.2 The James Webb Space Telescope ACS system designed in this chapter. FGS is a fine guidance sensor.

The design presented in this chapter uses the actuators and sensors selected for JWST. It does not use the same algorithms or control-system structure. Idealized models for all sensors and actuators are used.

24.1. Requirements

ACS requirements are summarized in Table 24.1 [1].

These requirements do not include jerk, the time derivative of acceleration, settling times, or jitter. Presumably, these were added in later iterations of the requirements.

Table 24.1 ACS requirements.

FGS is a fine guidance sensor.

Requirement	Time
90 degree slew	60 minutes
Pointing	7 arcsec
Pointing FGS	1 arcsec
Settling time	
Jitter	
Jerk	

24.2. Spacecraft model

The James Webb CAD model is shown in Fig. 24.3. The telescope mirror is on the right. The large structure in the middle is the sunshade. The spacecraft bus, consisting of all the hardware needed to run the spacecraft, is on the left. The panel on the right bottom is the momentum-balance panel. James Webb uses the axes designations J1, J2, and J3. (+J1,+J2,+J3) are (+X,+Y,+Z) in the diagram. Fig. 24.3 shows the James Webb Coordinate system.

James Webb Space Telescope parameters are given in Table 24.2.

SCAT stands for Secondary Combustion Augmentation Thrusters. These use an oxidizer with hydrazine. This gives them a higher exhaust velocity, thus lowering fuel consumption. The bias rate ensures that the wheels do not pass through zero. This prolongs the life of the wheels and eliminates noise due to commutation issues at zero wheel speed. Most wheels use trapezoidal commutation that produces noisy wheel-speed measurements at low speed.

The thrusters are a blowdown system, shown in Fig. 24.4, so the thrust will decrease with time. Typically, blowdown systems operate between 350 and 100 psi.

24.3. Disturbances

The spacecraft resides at the L2 Lagrange point. The only disturbances that impact the attitude are

1. Solar torques;
2. Thermal torques;
3. RF torques.

Solar torques are from the interaction of the Sun with the surfaces. Thermal torques are due to emissions from radiators or other hot objects on the surface of the spacecraft. RF torques are due to RF emissions from antennas. Gravity gradient, albedo, planetary radiation, residual dipole, and drag torques do not exist at L2.

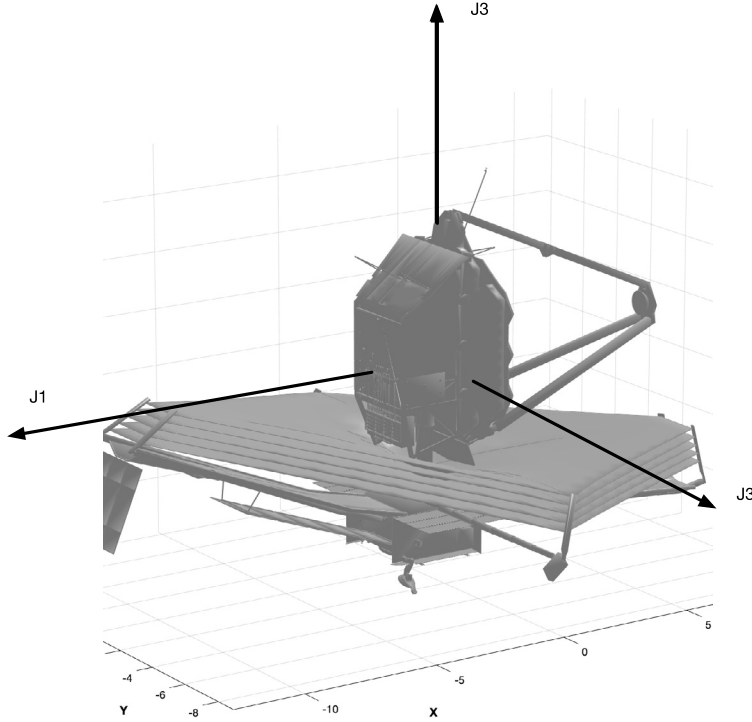


Figure 24.3 The James Webb Space Telescope CAD model. The surfaces are divided into triangles.

The thermal, solar, and RF torques are of the form

$$\mathbf{T} = (\mathbf{r}_k - \mathbf{r}_c) \times \mathbf{F}_k \quad (24.1)$$

where $\mathbf{r}_c$ is the center-of-mass and $\mathbf{r}_k$ is the location of the element producing the force $\mathbf{F}_k$. Solar forces are due to the Sun flux reflecting off the element. Fig. 24.5 shows the 2D layout.

The transmit power is quite low, 50 W, leading to a force of P/c (power divided by the speed-of-light) or $0.17 \mu\text{N}$. The offset from the center-of-mass is about 3 m so the disturbance torque is $0.5 \mu\text{N m}$. If we assume 9.5 hours of operation, the momentum build-up would be

$$\mathbf{h} = \mathbf{tr} \frac{\mathbf{P}}{c} \quad (24.2)$$

where $\mathbf{r}$ is the moment arm and t is the time of operation. The total momentum growth is 0.017 N m s , which is negligible.

Table 24.2 James Webb Space Telescope parameters [2] relevant to ACS. Principal axes are found primarily through a rotation about the y -axis. The center-of-mass is for the CAD representation.

Parameter	Value	Units
Inertia	$\begin{bmatrix} 67\,946 & -83 & 11\,129 \\ -83 & 90\,061 & 103 \\ 11\,129 & 103 & 45\,821 \end{bmatrix}$	kg m ²
Inertia (Principal Axes)	$\begin{bmatrix} 72\,575 & 0 & 0 \\ 0 & 90\,061 & 0 \\ 0 & 0 & 41\,191 \end{bmatrix}$	kg m ²
Center-of-mass	$\begin{bmatrix} 0.46 \\ 0 \\ 2.58 \end{bmatrix}$	m
Momentum-wheel torque	0.072	N m
Wheel spin-axis Inertia	0.1295	kg m ²
Wheel bias rate	2700	rpm
Wheel maximum rate	6000	rpm
Number of wheels	6	
ACS thrusters (16)	4.45	N
Delta-V (SCAT) thrusters (4)	35	N
Dinitrogen tetroxide [3]	133	kg
Hydrazine [3]	168	kg
RF Power [4]	50	W
Data transmit time [4]	9.5	hours

JWST has very sophisticated radiators for its optical system [6], and heat-rejection systems for the spacecraft bus.

The solar-disturbance model can be quite simple since the Sun is always on one side of the sunshield. This means a single plate, defined by an area, surface properties, center of pressure, and location, can be used. The sunshield is shown in Fig. 24.6.

The sunshield [7] is 21.197 m $\times$ 14.162 m. The outermost layer has a “doped-silicon” coating. Assume that the reflection coefficients are purely specular. The solar force for pure specular reflection is

$$F_k = -2a_k \frac{p}{c} (s^T n_k) n_k \quad (24.3)$$

where s is the vector to the Sun, a_k is the area, and n_k is the normal. The model is shown in Fig. 24.7.

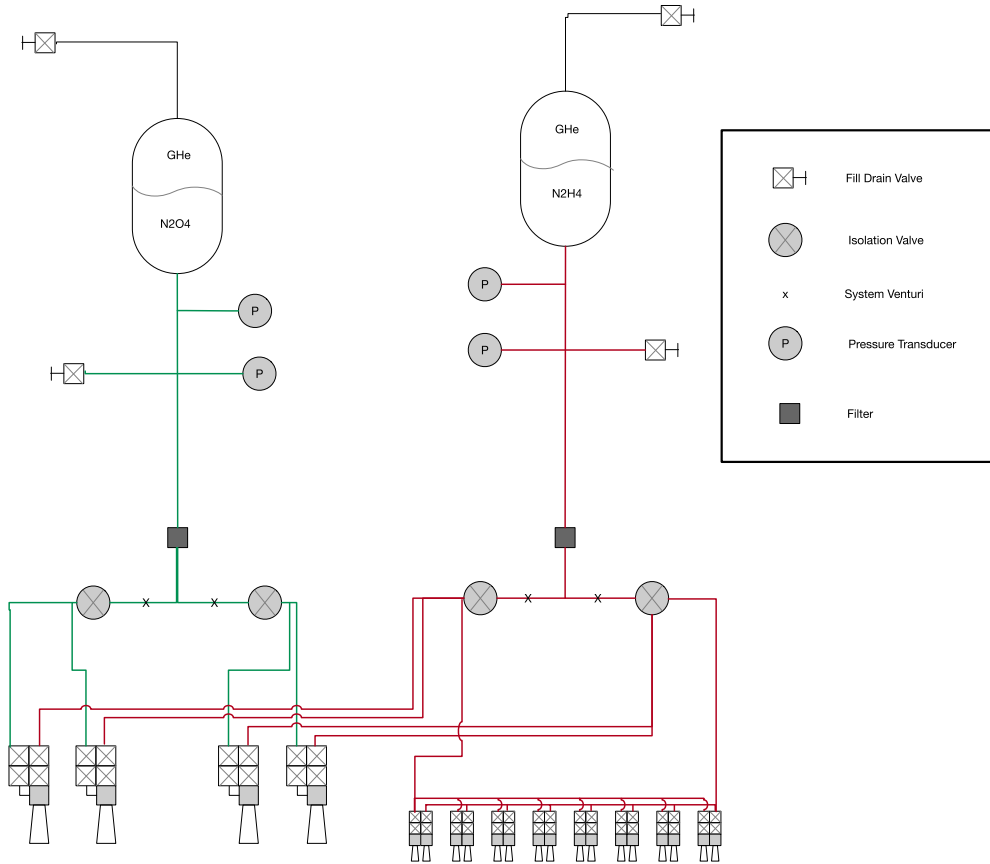


Figure 24.4 Propulsion system [5].

Example 24.1 gives a disturbance analysis. The center-of-mass is chosen so that the flap exactly balances the solar-pressure torques when the Sun is along the $-z$ -axis. The script shows the variation for slight γ angles.

24.4. Attitude maneuvers

The attitude-maneuver requirement is a 90-degree slew in 60 minutes. Considerable work has been done on slewing JWST [8,2]. For this analysis, we will assume there are no requirements on jerk, the time derivative of angular acceleration, so that we can use a bang-bang maneuver that is filtered to not excite slosh dynamics. For constant acceleration, the required torque is

$$T = \frac{4I\theta}{t^2} \quad (24.4)$$

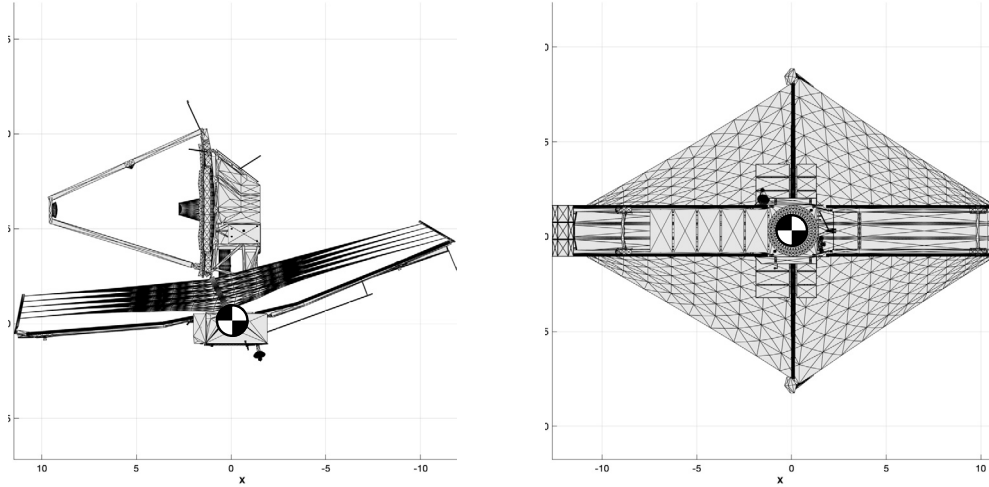


Figure 24.5 James Webb Space Telescope 2D drawing showing the approximate center-of-mass.



Figure 24.6 The James Webb Space Telescope sunshield. Image courtesy of NASA.

where t is the maneuver duration, I is the inertia, and θ is the maneuver angle. The maximum momentum stored for the maneuver is

$$h = \frac{Tt}{2} \quad (24.5)$$

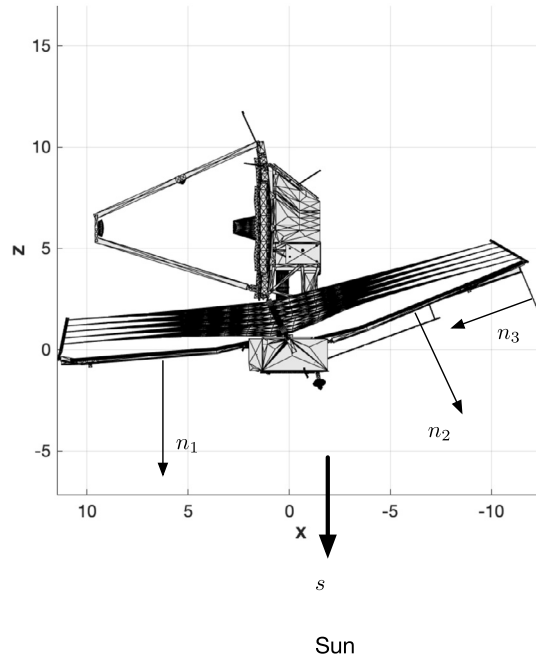


Figure 24.7 Disturbance model with three plates.

The largest inertia is I_{yy} in the principal axes, which is $90\,061 \text{ kg m}^2$. The torque is 0.044 N m and the required momentum storage is 78 N m s . The maximum momentum that can be stored in a wheel is 44.7 N m s . This means at least two wheels are required for a maneuver of this size.

24.5. Momentum control

Momentum control is done with thrusters and the momentum panel. Normally thruster unloading would be done with the 4.5-N thrusters. However, momentum can also be unloaded during station-keeping maneuvers using a combination of SCAT and ACS thrusters.

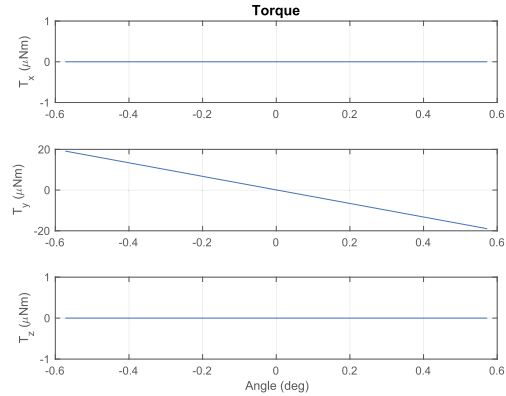
24.6. Attitude control

Reaction wheels are used for normal pointing control, including all attitude maneuvers. Thrusters are used for momentum unloading and also for attitude control during station-keeping maneuvers. Using the reaction wheels does not make much sense since the momentum gained during the maneuver would need to be unloaded with thrusters later.


```

1 %% Disturbance analysis
2
3 shade2Ang = 20*pi/180;% rad
4 flapAng   = 25*pi/180; % rad
5 yFlap     = 1.66+1.04;
6 lFlap     = sqrt((12.49-11.46)^2 + (3.82-1.48)
7           ^2);
8 aFlap     = yFlap*lFlap;
9 aShade    = 21.2*14.2;
10 cM        = [-0.5848;0;2.58];
11 p         = 1367/3e8;
12 s         = [0;0;-1];
13 n         = [[0;0;-1] [-sin(shade2Ang);0;-cos(
14           shade2Ang)] [cos(flapAng);0;-sin(flapAng)
15           ]];
16 a         = [aShade/2 aShade/2 aFlap];
17 r         = [[ 5;0;0.66] [-5;0;1.35]
18           [-12;0;2.47]];
19
20 % The CM with the sunshade
21 cM(1)     = fminsearch(@fun,-0.5,[],n,a,r,p,s,cM
22           );
23
24 % Shade 1, Shade 2, Flap
25 ang       = [-0.01 0 0.01];
26 torque    = zeros(3,length(ang));
27 for j = 1:length(ang)
28   s        = [sin(ang(j));0;-cos(ang(j))];
29   torque(:,j) = TorqueCalc(n,a,r,p,s,cM);
30 end
31
32 Plot2D(ang*180/pi,torque*1e6,'Angle_(deg)',...
33   {'T_x(\mu Nm)' 'T_y(\mu Nm)' 'T_z(\mu Nm)'},'
34   Torque');
35
36 function y = fun(x,n,a,r,p,s,cM)
37
38 cM(1) = x;
39 torque = TorqueCalc(n,a,r,p,s,cM);
40 y      = abs(torque(2));
41
42 end
43
44 function torque = TorqueCalc(n,a,r,p,s,cM)
45
46 torque = [0;0;0];
47 for k = 1:3
48   f      = -2*p*a(k)*(n(:,k)'*s)*n(:,k);
49   torque = torque + Cross(r(:,k)-cM,f);
50 end
51
52 end

```



Example 24.1: Disturbance torque as a function of Sun angle.

24.7. Torque distribution

Torque distribution uses linear programming. Reaction wheels produce variable torques based on the voltage applied to the motor. The thrusters are on/off, so pulsewidth modulation is used to generate “proportional” torques. Torque distribution uses linear programming. This is

$$x = Au \quad (24.6)$$

$$\min(c^T u) \quad (24.7)$$

$$u \geq 0 \quad (24.8)$$

The A matrix is a 3-by- n array of torque vectors giving the maximum vector of each torque. For the wheels, we add a positive and negative torquer-vector set. This allows the algorithm to produce negative reaction wheel commands. We put the six wheels at 45-degree angles with respect to the z -axis and oriented them in a hexagon around z . The thrusters are placed on the four sides of the bus. They are angled so that their plumes go in the $-z$ -direction. Fig. 24.8 shows the layout.

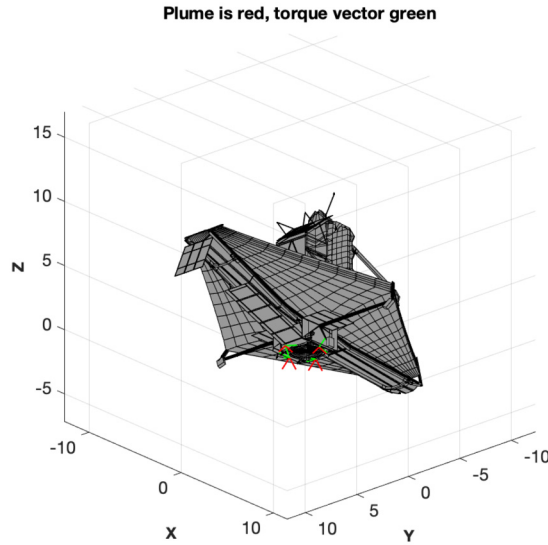


Figure 24.8 Thruster layout. There are eight thruster pairs in four locations at the bottom edge of the bus. The thrust vectors are all at 45 degrees with respect to the sunshade.

24.8. Attitude determination

Fig. 24.9 shows the star-camera-based attitude determination system.

It is useful to study the Kalman filter using a single-axis filter. The IMU model is

$$\dot{\theta}_i = \omega + b + \eta_{\theta_i} \quad (24.9)$$

$$\dot{b} = \eta_b \quad (24.10)$$

where ω is the true body rate. The IMU outputs an integrated rate. b is the bias, which is driven by a random-walk process. η_{θ_i} and η_b are Gaussian white noise. The IMU model is linear and the measurement is assumed to be the angle θ_b .

$$\gamma = \theta_b + \eta_{\theta_m} \quad (24.11)$$

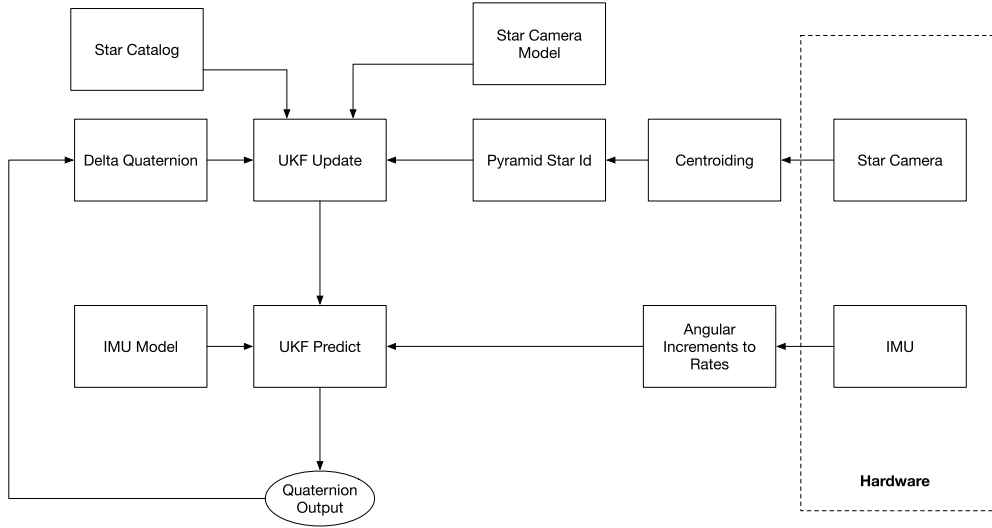


Figure 24.9 Attitude determination using an unscented Kalman filter (UKF).

where θ_b is the actual angle and η_{θ_m} the measurement noise. The model is in the form

$$\dot{x} = ax + bu \quad (24.12)$$

$$y = cx \quad (24.13)$$

$$a = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad (24.14)$$

$$b = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (24.15)$$

$$c = \begin{bmatrix} 1 & 0 \end{bmatrix} \quad (24.16)$$

$$x = \begin{bmatrix} \theta \\ b \end{bmatrix} \quad (24.17)$$

This single-axis model assumes that the star-camera measurement produces an angle as the product.

The model needs to be converted to discrete time for a standard Kalman filter. A standard Kalman filter will work for this single-axis, system. Note that

$$\dot{\theta}_b = \omega_b \quad (24.18)$$

The Kalman filter has two parts, state prediction and measurement incorporation. The Kalman-filter measurement incorporation is

$$k = \frac{ph^T}{hph^T + r} \quad (24.19)$$

$$x_e = k(\gamma - hx_e) \quad (24.20)$$

$$p = (E - kh)p \quad (24.21)$$

where E is a 2-by-2 identity matrix, p is the covariance matrix, r is the measurement covariance matrix, and the measurement matrix h is

$$h = \begin{bmatrix} 1 & 0 \end{bmatrix} \quad (24.22)$$

which is the same as the c matrix. The measurement is just θ_b . The prediction step is

$$x_e[k] = ax_e[k] + b \left(\frac{\theta_i[k] - \theta_i[k-1]}{\Delta t} \right) \quad (24.23)$$

$$p = apa^T + q \quad (24.24)$$

where a is the state-transition matrix, q is the model covariance matrix, and Δt is the time step. The output of the rate-integrating gyro is differenced to get the rate. This is where the IMU connects to the model. The IMU output is not a measurement.

Example 24.2 implements the Kalman filter. The script gives three choices for IMUs. The superior noise-filtering characteristics of the hemispherical resonating gyro are evident.

24.9. Simulation

This section shows a maneuver with a failure of a reaction wheel during a reorientation.

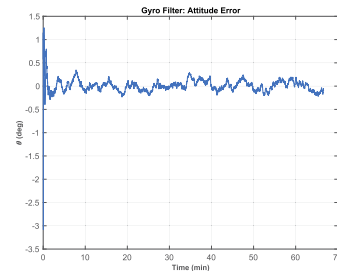
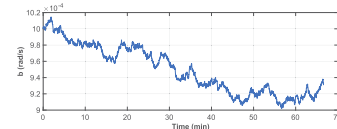
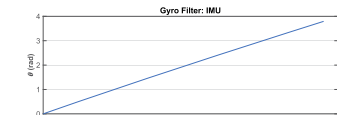
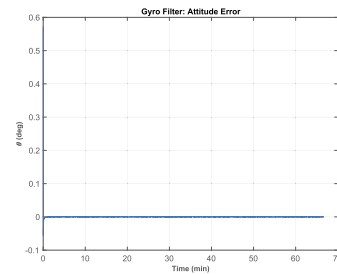
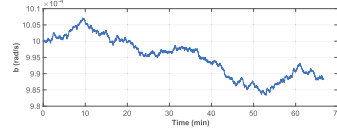
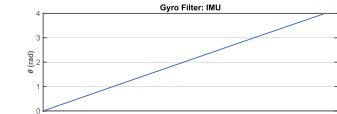
For James Webb, the major physical effects, beyond the rigid body are reaction wheels, slosh, and flexibility. The spacecraft has two fuel tanks, one for hydrazine and the other for the SCAT oxidizer. The spacecraft has six reaction wheels.

The simulation does not include slosh or flexibility. Six reaction wheels are employed to control the spacecraft using a PID controller. A momentum controller using thrusters is employed. The response is what would be expected from a PID controller. The pseudoinverse distributes the torque among the six wheels that are oriented in a pyramid. Fewer wheels could be used but they might not have sufficient momentum storage or torque capabilities. Six wheels also provide redundancy. The last plot shows the inertial angular momentum. Since all torques are internal, it is conserved.

```

1
2 model      = 'MEMS'; % 'adi' 'HRG', 'MEMS'
3 n          = 4000; % Number of time steps
4
5 %% IMU Model
6 h          = [1 0]; % Measurement matrix
7 a          = [0 1; 0 0];
8 b          = [1; 0];
9 dT         = 1;
10 [phi, gamma] = C2DZOH(a, b, dT);
11
12 %%
13 p          = 0.001*eye(2); % Set the initial
14             covariance
15 switch(lower(model))
16 case 'adi'
17     d = struct('etaB', RW2SDev(2), 'etaTheta', 0.75*
18               pi/180, 'u', 0, 'omega', 0);
19     sigmaY = pi/180;
20 case 'hrg'
21     d = struct('etaB', RW2SDev(0.001), 'etaTheta',
22               .60*0.02*4.85E-6, 'u', 0, 'omega', 0);
23     sigmaY = 4.85E-6; % 1 arc-second in radians
24 case 'mems' % CRS39A Series Silicon Sensing
25     d = struct('etaB', RW2SDev(0.006), 'etaTheta',
26               .006*pi/180, 'u', 0, 'omega', 0);
27     sigmaY = pi/180; % 1 arc-second in radians
28 end
29
30 r = sigmaY^2;
31 q = diag([d.etaTheta d.etaB].^2); % Plant noise
32   covariance
33
34 %% Simulation
35 x = [0; 0.001];
36 xE = [0.01; 0.005];
37 t = (0:n-1)*dT;
38
39 % For rate estimation from the angle
40 x1Old = 0;
41
42 % Run the filter for n steps
43 xP = zeros(7, n);
44 for j = 1:n
45     % True angle
46     theta = d.omega*t(j);
47
48     % Measurement
49     y = theta + sigmaY*randn;
50
51     % Plot storage
52     xP(:, j) = [x; xE; y; diag(p)];
53
54     % Measurement incorporation
55     k = psh'/(h*psh' + r);
56     xE = xE + k*(y - h*xE);
57
58     % Measurement update of covariance
59     p = (eye(2) - k*h)*p;
60
61     % State propagation Input is the gyro angle
62     xE = phi*xE + gamma*(x1Old - x(1))/dT;
63     p = phi*p*phi' + q;
64     x1Old = x(1);
65
66     % Integrate
67     x = RK4(@RIG, x, dT, 0, d);
68 end
69
70 theta = d.omega*t;
71 delta = (xP(3,:) - theta)*180/pi;
72
73 leg = {'True' 'Estimated'};
74 yL = {'\theta_0(rad)', 'b_0(rad/s)', ...
75       '\theta_E_0(rad)', 'b_E_0(rad/s)', ...
76       'y_0(rad)', 'p_theta', 'p_b'};
77 TimeHistory(t, xP(1:2,:), yL(1:2), 'IMU');
78 TimeHistory(t, xP(6:7,:), yL(6:7), 'Covariance');
79 TimeHistory(t, delta, ['\theta(deg)', 'Error']);
80
81 %% Gyro RHS
82 function xDot = RIG(x, ~, d)
83
84 b = x(2);
85 bDot = d.etaB*randn;
86 thetaDot = d.omega + b + d.etaTheta*randn + d.u;
87 xDot = [thetaDot; bDot];
88 end

```

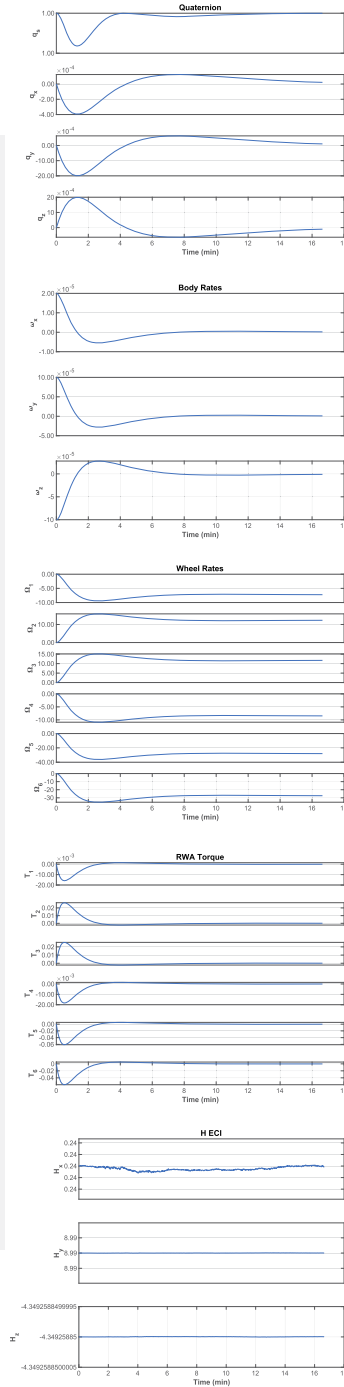


Example 24.2: Single-axis Kalman filter. The first two plots are for the HRG. The second two for a good-quality MEMS IMU.

```

1 %% Constants
2 secInDay = 86400;
3 controlOn = true;
4 mmOn = true; % Momentum management
5
6 %% User initialization
7 n = 1000; % Number of simulation steps
8 dT = 1.0; % Time step (s)
9 inertia = [67946 -83 11129; -83 90061 103; 11129
10           103 45821];
11
12 qTarget = [1;0;0;0];
13
14 % Initial Julian date
15 JD0 = Date2JD([2023 4 4]);
16
17 % Initial states
18 q = [1;0;0;0];
19 omega = [0;0;0]; % rad/s
20 dOmega = [0.00002;0.0001;-0.0001]; % Delta
    angular rate
21 omegaRWA = [0;0;0;0;0;0]; % rad/s
22 torqueD = [2;0;0]*1e-6; % Disturbance torque
23
24 %% State vector
25 x = [q; omega+dOmega; omegaRWA];
26 dRHS = RHSJWST;
27 dRHS.inertia = inertia;
28
29 % Simulate
30 xP = zeros(25,n);
31
32 % Edit the control system data structure
33 dC = ACSJWST;
34 dC.dPID.q_desired_state = qTarget;
35
36 dC.dT = dT;
37 dC.dRWA = dRHS.dRWA;
38 dC.mmOn = mmOn;
39 dC.dPID.inertia = dRHS.inertia;
40
41 % Simulation loop
42 for k = 1:n
43     if (controlOn)
44         [dC,tThruster,dRWA] = ACSJWST(x,dC);
45     else
46         tThruster = [0;0;0]; dRWA = [0;0;0;0;0;0];
47     end
48     dRHS.torque = tThruster + torqueD;
49     dRHS.dRWA = dRWA;
50     [~,hECI] = RHSJWST(x,0,dRHS);
51     xP(:,k) = [x;tThruster;dRWA;hECI];
52     x = RK4(@RHSJWST,x,dT,0,dRHS);
53 end
54
55 % Plotting
56 t = (0:n-1)*dT;
57
58 yL = {'q_s' 'q_x' 'q_y' 'q_z' ...
59       '\omega_x' '\omega_y' '\omega_z' ...
60       '\Omega_1' '\Omega_2' '\Omega_3' ...
61       '\Omega_4' '\Omega_5' '\Omega_6' ...
62       'T_x' 'T_y' 'T_z' ...
63       'T_1' 'T_2' 'T_3' 'T_4' 'T_5' 'T_6' ...
64       'H_x' 'H_y' 'H_z'};
65
66 iL = {'Quaternion' 'Body_Rates' 'Wheel_Rates' ...
67       'REA_Torque' 'RWA_Torque' 'H_ECI'};
68
69 kL = {1:4 5:7 8:13 14:16 17:22 23:25};
70
71 for j = 1:length(iL)
72     k = kL{j};
73     TimeHistory(t,xP(k,:),yL(k),iL{j});
74 end
75
76 Figui

```



Example 24.3: JWST simulation. The PID controller damps rates using the six reaction wheels.

References

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